

JINTONG LIU

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EDUCATION

Ph.D., Computer Science, The Ohio State University 08/2024 – now
M.E., Computer System Architecture, Huazhong Univ. of Sci. and Tech. 09/2019 – 06/2022
B.E., Electronic Science and Technology, Huazhong Univ. of Sci. and Tech. 09/2015 – 06/2019

RESEARCH INTEREST

Distributed systems, performance analysis, fault tolerance, databases.

PUBLICATIONS

- **Jintong Liu**, Shenggang Wan, Xubin He. Alias-Chain: Improving Blockchain Scalability via Exploring Content Locality among Transactions, in: 36th IEEE International Parallel and Distributed Processing Symposium (**IPDPS '22**), Lyon, France, 30 May – 3 June 2022, IEEE.
- Chenglong Yi*, **Jintong Liu***, Shenggang Wan, Juntao Fang, Bin Sun, Liqiang Zhang and Changsheng Xie. Data Deduplication Based on Content Locality of Transactions to Enhance Blockchain Scalability. Transactions on Architecture and Code Optimization (**TACO**), ACM.

EXPERIENCE

Proj: Enhancing Timeout with Causal Tracing 02/2025 – Current

- PI: Dr. Yang Wang
- Leverage tracing in various distributed systems through graph mining to enhance timeout mechanism.

Blockchain Storage Research Center 09/2019 – 06/2022

- PI: Dr. Shenggang Wan
- Investigated different blockchains' architecture, consensus, cryptography, smart contract, etc.
- Implemented a simulator for blockchain performance measurement and analysis.
- Proposed a data deduplication design to scale out blockchains.
- Designed and conducted experiments on a 100-machine cluster.
- Oral presentation at the IPDPS '22 conference.

Memolabs

Research Intern 09/2021 – 03/2022

- Implemented consensus *hotstuff* and related modules, introduced lightweight optimizations to address some liveness problems in certain specific scenarios within the paper version protocol, which can achieve higher performance and non-rollback manner.
- Designed and implemented interfaces of *Sparse Merkle Tree* for state transition and validation in state machines, optimized SMT's key-value structure to support the safe multiversion feature.

Software Engineer 06/2022 – 05/2024

- Developed the product version of the *Alias-Chain* in Go, as an integrated optimized module in Ethereum, which enabled on-chain data shrinkage.
- Designed and implemented *Indexer* for monitoring blockchains' states and specified smart contracts.
- Implemented *Decentralized Profile* to store users' profiles on-chain and off-chain by using smart contracts and decentralized storage network.
- Deployed *Reverse Image Search* as a service for decentralized storage networks.
- Designed and implemented part of the *Decentralized Computing Services* by smart contracts and kubernetes for computing resource allocation.

SKILLS

- Paxos, Raft, PBFT-like
- Cassandra, BadgerDB, MongoDB, Kubernetes
- Java, Python, Go, C/C++, SQL/NoSQL, LaTeX